

JEE MAIN REVISION MASTER: SURFACE CHEMISTRY OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 113 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Surface Chemistry — Standard Calculations & Problem Archetypes	57 PYQs	Quantitative Analysis
Very High	Surface Chemistry — Applied Cases, Reactions & Traps	42 PYQs	Applied Topics
Very High	Surface Chemistry — Core Principles & Governing Laws	20 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Surface Chemistry — Standard Calculations & Problem Archetypes	4 g. (Given : Molar mass of acetic acid = 60 g mol, $\frac{x}{m}$), $\frac{x}{m}$
Surface Chemistry — Applied Cases, Reactions & Traps	20-40kJmol ⁻¹ , $\frac{x}{m}=kp^{0.5}$, $\frac{x}{m}=kP^{1/n}$
Surface Chemistry — Core Principles & Governing Laws	ΔH_{ads} , $Fe_2O_3.xH_2O$, w_1

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Surface Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 57 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Surface Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Evening Shift] Q#1 (MCQ): "The adsorbent used in adsorption chromatography is/are - A. silica gel B. alumina C. quick lime D. magnesia Choose the most appropriate ans..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Four gases A, B, C and D have critical temperatures 5.3, 33.2, 126.0 and 154.3 K respectively. For their adsorption on a fixed amount of ch..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "In figure, a straight line is given for Freundlich Adsorption ($y = 3x + 2.505$). The value of $(1/n)$ and $\log K$ are respectively...." [Ans: N/A]

3.2 Surface Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 42 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Surface Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (Numerical): "Based on the given figure, the number of correct statement/s is/are _____. A. Surface tension is the outcome of equal attractive and re..." [Ans: 2]
- [2023 Evening Shift] Q#2 (MCQ): "Ferric chloride is applied to stop bleeding because -..." [Ans: D]
- [2023 Morning Shift] Q#2 (MCQ): "Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R Assertion A: Amongst He, Ne, Ar and Kr..." [Ans: N/A]

3.3 Surface Chemistry — Core Principles & Governing Laws (Very High Yield — 20 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Surface Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Surface Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Surface Chemistry PYQs.

Representative Source PYQs:

- [2024 Morning Shift] Q#2 (MCQ): "'Adsorption' principle is used for which of the following purification method?..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "The enthalpy change for the adsorption process and micelle formation respectively are..." [Ans: N/A]
- [2023 Morning Shift] Q#3 (MCQ): "Which of the following salt solutions would coagulate the colloid solution formed when FeCl_3 is added to NaOH solution, at the fastest..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 12th April Morning Shift (Q#1)

Four gases A, B, C and D have critical temperatures 5.3, 33.2, 126.0 and 154.3 K respectively. For their adsorption on a fixed amount of charcoal, the correct order is:

- (A) $A > C > B > D$
- (B) $B > D > A > C$
- (C) $C > A > B > D$
- (D) $D > B > A > C$

Correct Answer: None

Solution Analysis: The extent of adsorption of a gas on a solid surface depends on the critical temperature of the gas and the temperature of the solid surface. The critical temperature is the temperature above which a gas cannot be liquefied, no matter how much pressure is applied. Therefore, gases with higher critical temperatures are less likely to be adsorbed on a solid surface than gases with lower critical temperatures. In this case, the gases are A, B, C, and D, with critical temperatures of 5.3 K, 33.2 K, 126.0 K, and 154.3 K, respectively. Therefore, the correct order of adsorption on a fixed amount of

Benchmark Problem 2: JEE Main 2023 (Online) 25th January Evening Shift (Q#1)

Based on the given figure, the number of correct statement/s is/are _____. A. Surface tension is the outcome of equal attractive and repulsive forces acting on the liquid molecule in bulk. B. Surface tension is due to uneven forces acting on the molecules present on the surface. C. The molecule in the bulk can never come to the liquid surface. D. The molecules on the surface are responsible for vapour pressure if the system is a closed system.

Correct Answer: 2

Solution Analysis: The correct statements are (B) Surface tension is due to uneven forces acting on the molecules present on the surface (D) The molecules on the surface are responsible for vapor pressure if the system is a closed system

Benchmark Problem 3: JEE Main 2023 (Online) 10th April Morning Shift (Q#3)

The enthalpy change for the adsorption process and micelle formation respectively are

- (A) $\Delta H_{\text{ads}} > 0$ and $\Delta H_{\text{mic}} < 0$
- (B) $\Delta H_{\text{ads}} > 0$ and $\Delta H_{\text{mic}} > 0$
- (C) $\Delta H_{\text{ads}} < 0$ and $\Delta H_{\text{mic}} < 0$
- (D) $\Delta H_{\text{ads}} < 0$ and $\Delta H_{\text{mic}} > 0$

Correct Answer: None

Solution Analysis: Adsorption is the process of a substance (adsorbate) being attracted to a surface (adsorbent). The enthalpy change for adsorption, ΔH_{ads} , is negative because the adsorbent and adsorbate are brought closer together, which releases energy. Micelle formation is the process of surfactant molecules clustering together in water to form micelles. The micelles have a hydrophobic (water-repelling) core and a hydrophilic (water-loving) surface. The enthalpy change for micelle formation, ΔH_{mic} , is positive because the surfactant molecules are moving from a state of high entropy (d